

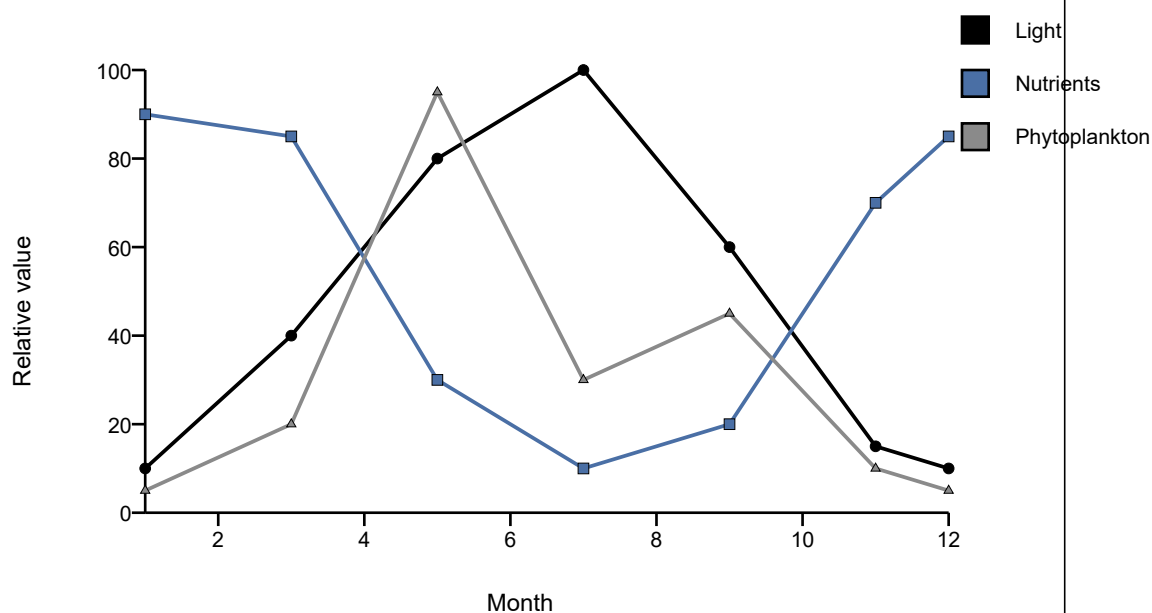
Questions by topic: Marine productivity & fishing - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 1 . 1

Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

0 1 . 2

Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

0 1 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 1 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

0 2

Table 1 shows features of some fishing methods.

0 2 . 1

Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage
Demersal trawling		
Purse seining		
Pelagic long-lining		

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Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

0	2	.	3
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Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

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0 3

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 3**1**

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

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0 3**2**

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

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0 3**3**

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

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03 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

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04

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

04 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer

thousand tonnes

04 . 2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer thousand tonnes

0 4 . 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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0 4 . 4

Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort

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Mean fish size and mean age

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0 5 . 1 Explain how each of the following reduces the environmental impact of fishing. **[8 marks]**

Square-mesh escape panels

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Minimum and maximum catchable sizes

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No-take zones

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Closed seasons

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0 5 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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06 . 1

Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

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06 . 2

Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

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06 . 3

Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

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06 . 4

Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

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07 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

07 . **1** Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer %

07 . **2** Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

07 . **3** Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

07 . **4** Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

[illegible]

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END OF QUESTIONS